Portfolio Optimization Under Uncertainty

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Independent work carried out outside the academic curriculum. All code is written in Python, is hosted in a GitHub repository and runs in Google Colab (see Section 20).

Abstract

This study asks whether sophisticated portfolio optimization delivers better out-of-sample, risk-adjusted performance than a simple equally weighted portfolio once estimation uncertainty, downside risk, transaction costs and changing market conditions are taken into account. Four strategies are compared (Equal Weight, Minimum Variance, Maximum Sharpe and Risk Parity) on a fixed universe of 20 EUR-denominated European equities, using a point-in-time walk-forward procedure with a 756-day training window, quarterly (63-day) rebalancing, a 10% weight cap and 10 basis points of transaction cost per unit of turnover. The evaluation period runs from January 2018 to October 2026 (2,192 daily observations). Maximum Sharpe is the descriptive leader, with an annualized return of 14.07%, volatility of 18.14% and a Sharpe ratio of 0.817, against 12.65%, 18.82% and 0.728 for Equal Weight. However, a paired HAC (Newey–West) test of the difference in mean daily returns gives $p = 0.621$, so the advantage cannot be distinguished from sampling variation. Maximum Sharpe also hits the 10% cap at every rebalance, holds about 10 effective assets instead of 20, and trades roughly 3.4 times as much as Equal Weight. The evidence supports a qualified conclusion: optimization can improve historical outcomes, but it does not demonstrably dominate a naive benchmark.

Keywords: portfolio optimization; Markowitz; maximum Sharpe; risk parity; Ledoit–Wolf shrinkage; estimation risk; walk-forward backtest; HAC inference; transaction costs.

1. Introduction

Portfolio optimization turns estimates of expected returns and covariances into portfolio weights. In theory, the optimizer exploits all available information to build the best possible trade-off between risk and return. In practice, those inputs are estimated with error, and an optimizer tends to magnify that error: it overweights assets whose estimated return is unusually high, or whose estimated risk is unusually low, which is often a sign of noise rather than genuine opportunity. The result can be unstable, concentrated portfolios that look excellent in-sample but disappoint out-of-sample.

A natural benchmark in this debate is the equally weighted portfolio, which requires no estimation at all. The question studied here is therefore:

> **Research question.** Does sophisticated portfolio optimization provide superior risk-adjusted performance compared with a simple equally weighted portfolio once estimation uncertainty, downside risk, transaction costs and changing market conditions are taken into account?

To answer it, the study combines portfolio theory, statistical inference, risk measurement, regime analysis and implementation diagnostics in one reproducible pipeline. It makes four contributions: (i) a strictly out-of-sample comparison of four portfolio construction rules on a European large-cap universe; (ii) formal inference on return differences using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors rather than relying on point estimates; (iii) a diagnosis of concentration, turnover and transaction costs, which determine whether paper performance can be implemented; and (iv) robustness checks on covariance estimation, return-estimate shrinkage and cost assumptions.

How the analysis fits together. The study is organized as a chain in which each step answers a question raised by the previous one. The data analysis (Section 4) identifies the estimation problems that optimizers will face. Portfolio construction (Section 5) shows what optimization promises in-sample. The out-of-sample results (Section 6) show how much of that promise is delivered. The dynamic risk analysis (Section 7) checks whether the result is stable over time, and the HAC inference (Section 8) whether it is larger than chance. The market regression (Section 9) and the regime analysis (Section 10) explain where the performance comes from and when. Concentration, turnover and costs (Sections 11, 12 and 14) show what it costs to obtain it, and the estimation robustness checks (Section 13) test whether it depends on one estimator. Section 16 then combines all of this into a direct answer to the research question. Table 1 maps each part of the research question to the analysis that addresses it.

Table 1. From the research question to the analyses.

Part of the research question	Question asked	Analysis (sections)	Why it is needed
Superior risk-adjusted performance	Do optimized portfolios beat Equal Weight out-of-sample?	Baseline results, wealth paths (6)	Core evidence; defines what superior means
Statistical reliability	Is the difference larger than chance?	HAC tests (8)	Separates a real effect from sampling noise
Source of performance	Is it alpha or just market risk?	Market regression (9)	Controls for market exposure
Changing market conditions	Does it depend on the volatility regime?	Dynamic risk, regimes (7, 10)	Tests stability across environments
Downside risk	Are tail losses smaller?	VaR, CVaR, drawdowns (6, 7)	Variance alone ignores tail losses
Estimation uncertainty	Does the result survive noisy inputs?	Data structure, concentration, shrinkage (4, 11, 13)	Optimizers amplify estimation error
Transaction costs	Can it be implemented?	Turnover, cost sensitivity (12, 14)	Paper returns are not net returns

The paper is organized as follows. Section 2 reviews the theoretical background. Section 3 describes the data, methodology and the mathematical definitions of all metrics. Sections 4 and 5 present exploratory analysis and portfolio construction. Sections 6 to 14 report out-of-sample results, risk dynamics, inference, regression, regime, implementation and robustness analyses. Sections 15 and 16 discuss the findings and give the answer to the research question. Sections 17 to 19 present limitations, directions for future research and conclusions, and Section 20 documents the code.

2. Theoretical Background

Modern portfolio theory starts with Markowitz (1952), who showed that investors should evaluate assets by their contribution to portfolio risk and return jointly, giving rise to the mean–variance efficient frontier.

Sharpe (1964) linked this framework to equilibrium asset pricing, and the Sharpe ratio, the excess return per unit of volatility, became the standard measure of risk-adjusted performance. The Maximum Sharpe portfolio is the point on the frontier with the highest ratio.

The practical weakness of mean–variance optimization is estimation risk. DeMiguel, Garlappi and Uppal (2009) compared a wide range of optimal portfolio models with the naive $1/N$ rule and found that none consistently beat it out-of-sample, mainly because of estimation error. Two families of remedies exist. The first improves the inputs: Ledoit and Wolf (2004) proposed shrinking the sample covariance matrix toward a structured target, which reduces its estimation noise, and expected returns can similarly be shrunk toward a common mean. The second avoids the most error-prone input: Minimum Variance requires only the covariance matrix, and Risk Parity (Maillard, Roncalli and Teïletche, 2010) requires only that each asset contribute an equal share of portfolio risk.

Because variance treats gains and losses symmetrically, the study also tracks downside measures such as the Sortino ratio, maximum drawdown, Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR; Rockafellar and Uryasev, 2000). Finally, because daily returns exhibit volatility clustering and some serial dependence, differences between strategies are tested with Newey–West (1987) HAC standard errors.

3. Data and Methodology

3.1 Data

The investment universe consists of 20 large-cap European stocks listed in Amsterdam, Frankfurt, Paris and Milan: ASML.AS, SAP.DE, SIE.DE, ALV.DE, DTE.DE, BAS.DE, ADS.DE, AIR.PA, MC.PA, OR.PA, SAN.PA, BNP.PA, SU.PA, AI.PA, ENEL.MI, ENI.MI, ISP.MI, UCG.MI, BMW.DE and RWE.DE. Daily data are downloaded from Yahoo Finance using the yfinance library. The requested start date is 1 January 2015, which provides the history needed for the first training window, and the last observation is 2 October 2026. The out-of-sample period runs from 5 January 2018 to 2 October 2026 and contains 2,192 daily observations per strategy. A broad European index proxy (EXSA.DE) is used for the market regression in Section 9.

3.2 Walk-forward design

At every rebalancing date, the previous 756 trading days (about three years) form the training sample, and portfolio weights are estimated using only information available at that date. The portfolio then drifts with market returns (buy-and-hold) for the next 63 trading days (about one quarter), after which the procedure is repeated. No future observation ever enters an estimate, so the reported results are genuinely out-of-sample. Table 2 summarizes the design.

3.3 Portfolio strategies

For N assets with weight vector $\mathbf{w}$ and daily return vector $\mathbf{r}_t$, the portfolio return and the budget constraint are:

$$R_{p,t} = \mathbf{w}'\mathbf{r}_t \tag{1}$$

$$\sum_i w_i = 1, \qquad 0 \le w_i \le 0.10 \tag{2}$$

Expected returns and the covariance matrix are estimated on the training window of length T:

$$\hat{\mu} = \frac{1}{T}\sum_t \mathbf{r}_t \tag{3}$$

$$\hat{\Sigma} = \frac{1}{T-1}\sum_t (\mathbf{r}_t - \hat{\mu})(\mathbf{r}_t - \hat{\mu})' \tag{4}$$

Table 2. Summary of the experimental design.

Parameter	Specification
Universe	20 EUR-denominated European equities
Data source	Yahoo Finance (yfinance), daily
Out-of-sample period	2018-01-05 to 2026-10-02 (2,192 daily observations)
Training window	756 trading days
Rebalancing frequency	63 trading days, buy-and-hold drift in between
Constraints	Long-only, fully invested, 10% maximum weight
Transaction cost	10 basis points per unit of turnover
Annualization / risk-free rate	252 periods per year / zero daily risk-free rate
Strategies	Equal Weight, Minimum Variance, Maximum Sharpe, Risk Parity

Equal Weight holds $1/N$ in every stock and uses no estimates. Minimum Variance chooses the weights that minimize portfolio variance, and uses only the covariance matrix:

$$\min_{\mathbf{w}} \ \mathbf{w}'\hat{\Sigma}\mathbf{w} \tag{5}$$

Repeating this minimization for a grid of target returns μ^* (with the same constraints) traces the efficient frontier shown in Figure 6:

$$\min_{\mathbf{w}} \ \mathbf{w}'\hat{\Sigma}\mathbf{w} \quad \text{subject to} \quad \mathbf{w}'\hat{\mu} = \mu^* \tag{6}$$

Maximum Sharpe maximizes the ex-ante Sharpe ratio and therefore relies on both estimated returns and covariances, which makes it the most estimation-intensive strategy:

$$\max_{\mathbf{w}} \ \frac{\mathbf{w}'\hat{\mu} - r_f}{\sqrt{\mathbf{w}'\hat{\Sigma}\mathbf{w}}} \tag{7}$$

Risk Parity targets approximately equal risk contributions across assets, where the contribution of asset i is:

$$RC_i = \frac{w_i(\hat{\Sigma}\mathbf{w})_i}{\sigma_p}, \qquad \sigma_p = \sqrt{\mathbf{w}'\hat{\Sigma}\mathbf{w}} \tag{8}$$

By construction the contributions add up to total portfolio risk, so equal risk contributions means that each asset accounts for the same share of it:

$$\sum_i RC_i = \sigma_p, \qquad \text{target: } RC_i = \sigma_p/N \tag{9}$$

3.4 Performance, risk and implementation metrics

All metrics are computed on the daily out-of-sample returns R_t of each strategy, with $n = 2{,}192$ observations and 252 trading days per year. Between rebalancing dates the weights drift with market returns, and on each rebalancing date k the transaction cost is deducted from the return:

$$w_{i,t+1} = \frac{w_{i,t}(1 + r_{i,t})}{1 + R_{p,t}} \tag{10}$$

$$R^{\text{net}}_{p,t} = R_{p,t} - TC_k \quad \text{on rebalancing days} \tag{11}$$

Return and volatility. The annualized return compounds the daily returns, and annualized volatility scales the daily standard deviation by the square root of 252:

$$R_{\text{ann}} = \Big(\prod_{t=1}^{n}(1 + R_t)\Big)^{252/n} - 1 \tag{12}$$

$$\sigma_{\text{ann}} = \sqrt{252}\, s_R, \qquad s_R^2 = \frac{1}{n-1}\sum_t (R_t - \bar{R})^2 \tag{13}$$

Risk-adjusted ratios. The Sharpe ratio divides the mean excess return by its volatility. The Sortino ratio replaces total volatility by downside deviation σ_d, so that only returns below the threshold are penalized:

$$SR = \sqrt{252}\,\frac{\bar{R} - r_f}{s_R} \tag{14}$$

$$\text{Sortino} = \sqrt{252}\,\frac{\bar{R} - r_f}{\sigma_d}, \qquad \sigma_d = \sqrt{\frac{1}{n}\sum_t \big[\min(R_t - r_f, 0)\big]^2} \tag{15}$$

Drawdown measures. With W_t the cumulative wealth, the drawdown D_t is the percentage fall from the previous peak, the maximum drawdown (MDD) is its worst value, and the Calmar ratio compares annualized return with that worst loss:

$$D_t = \frac{W_t}{\max_{s \le t} W_s} - 1, \qquad \text{MDD} = \min_t D_t \tag{16}$$

$$\text{Calmar} = \frac{R_{\text{ann}}}{|\text{MDD}|} \tag{17}$$

Tail risk. The 95% Value-at-Risk is the daily loss exceeded on only 5% of days, and the Conditional Value-at-Risk is the average loss on those worst days. Both are reported as positive numbers:

$$\text{VaR}_{95} = -Q_{0.05}(R_t), \qquad \text{CVaR}_{95} = -E\big[R_t \mid R_t \le -\text{VaR}_{95}\big] \tag{18}$$

Concentration and turnover. The effective number of assets is the inverse of the Herfindahl index of the weights: it equals N for equal weights and falls as the portfolio concentrates. Turnover measures how much is traded at each rebalance (relative to the drifted weights), and the cost is proportional to it:

$$N_{\text{eff}} = \frac{1}{\sum_i w_i^2} \tag{19}$$

$$TO_k = \sum_i \big| w_{i,k} - w_{i,k}^{\text{drift}} \big| \tag{20}$$

$$TC_k = 0.001 \times TO_k \tag{21}$$

3.5 Statistical inference

To test whether a strategy s outperforms Equal Weight, the daily return difference $\Delta R_t = R_{s,t} - R_{\text{EW},t}$ is computed and its mean is tested with a Newey–West HAC standard error, which is robust to serial correlation and heteroskedasticity:

$$t = \frac{\overline{\Delta R}}{SE_{\text{HAC}}(\overline{\Delta R})} \tag{22}$$

The HAC variance of the mean difference uses L autocovariances $\hat{\gamma}_l$ of ΔR_t with Bartlett weights (Newey and West, 1987), so that the standard error remains valid when daily differences are serially correlated:

$$\hat{V}_{\text{HAC}} = \hat{\gamma}_0 + 2\sum_{l=1}^{L}\Big(1 - \frac{l}{L+1}\Big)\hat{\gamma}_l, \qquad SE_{\text{HAC}} = \sqrt{\hat{V}_{\text{HAC}}/n} \tag{23}$$

The null hypothesis is that the mean daily return difference is zero. Market exposure is examined with a CAPM-style regression of each strategy on the market proxy:

$$R_{s,t} = \alpha_s + \beta_s R_{m,t} + \varepsilon_{s,t} \tag{24}$$

For readability, daily quantities are converted to annual terms by simple multiplication, which is the approximation used in the text for mean differences and alphas:

$$\overline{\Delta R}_{\text{ann}} \approx 252 \times \overline{\Delta R}, \qquad \alpha_{\text{ann}} \approx 252 \times \alpha \tag{25}$$

3.6 Covariance structure and shrinkage

The exploratory analysis uses the correlation matrix and the eigenvalues of the sample covariance matrix. The first measures co-movement between pairs of stocks; the second decomposes total variance into orthogonal directions (Q contains the eigenvectors and Λ the eigenvalues), the largest of which represents the common market factor:

$$\rho_{ij} = \frac{\hat{\Sigma}_{ij}}{\hat{\sigma}_i \hat{\sigma}_j} \tag{26}$$

$$\hat{\Sigma} = Q\Lambda Q', \qquad \Lambda = \mathrm{diag}(\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N) \tag{27}$$

Because the smallest eigenvalues are the most poorly estimated, Ledoit and Wolf (2004) shrink the sample covariance matrix toward a structured target F (in common implementations, a scaled identity matrix), with an intensity δ chosen to minimize the expected estimation error. Expected returns are shrunk in a similar spirit toward their cross-sectional average (Section 13):

$$\hat{\Sigma}_{LW} = \delta F + (1-\delta)\hat{\Sigma}, \qquad 0 \leq \delta \leq 1 \tag{28}$$

3.7 Volatility regimes

For the regime analysis, rolling 63-day annualized volatility of the Equal Weight portfolio is computed, and each day is assigned to a regime according to the quantiles of this series ($Q_{0.20}$ and $Q_{0.80}$):

$$\sigma_t^{\mathrm{roll}} = \sqrt{252} \cdot s(R_{t-62}, \ldots, R_t) \tag{29}$$

$$\mathrm{Regime}_t = \text{low if } \sigma_t^{\mathrm{roll}} \leq Q_{0.20};\ \text{ high if } \sigma_t^{\mathrm{roll}} \geq Q_{0.80};\ \text{ normal otherwise} \tag{30}$$

4. Exploratory Data Analysis

Why this analysis. An optimizer can only be as good as the estimates it is given, and those estimates come from historical data. Before comparing strategies we therefore study the data itself, to learn which estimation problems the optimizers will face. Four properties matter: how unequal past returns were (Figure 1), how much diversification is really available (Figure 2), whether returns are normal or have heavy tails (Figures 3 and 4), and how concentrated the risk is in a few directions (Figure 5). Each of these produces a prediction that later sections test.

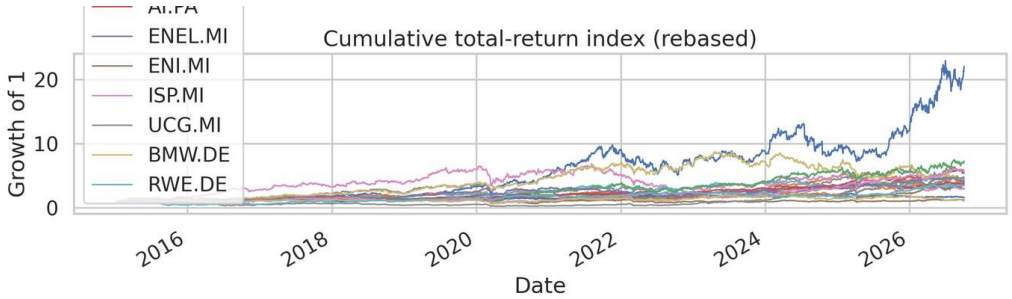


Figure 1. Cumulative total-return index of the 20 stocks (rebased to 1).

Commentary. The figure rebases every stock to 1 at the start of the sample. Dispersion across stocks is very large: most stocks multiply their value by roughly 2 to 7 over the period, while one stock reaches about 20 times its starting value, mostly through a strong rally from 2024 onward. Such differences matter for optimization. An estimator of expected returns based on a three-year history will tend to favour recent winners, and if these winners then fail to repeat, a strategy that chases them (as Maximum Sharpe does) can be hurt. This is a first, visual illustration of estimation risk.

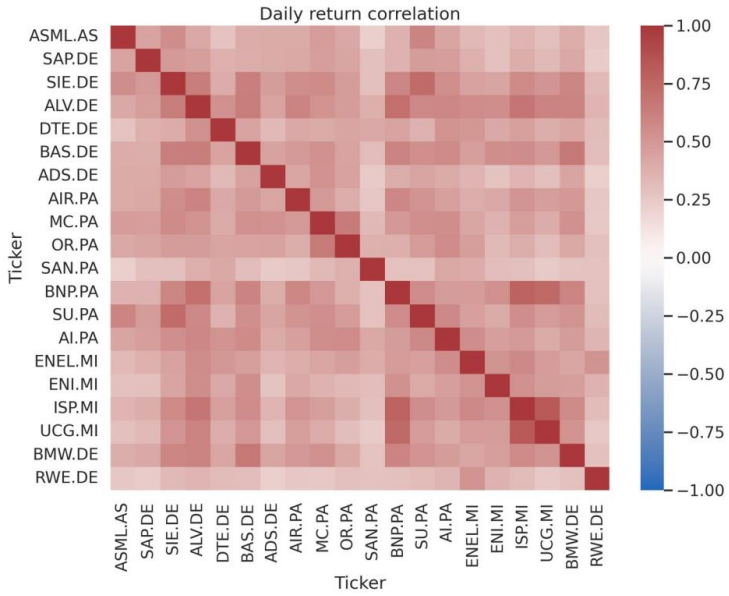


Figure 2. Correlation matrix of daily returns.

Commentary. All pairwise correlations are positive, and the majority lie roughly between 0.25 and 0.60, which indicates a strong common European market component. The highest correlations occur among the banks (BNP.PA, ISP.MI and UCG.MI), reflecting their shared sensitivity to interest rates and credit conditions. Stocks such as SAN.PA, DTE.DE and RWE.DE show visibly lower correlations with most of the universe, so they are natural candidates for diversification and are the kind of assets a variance-minimizing optimizer will favour. Overall, diversification gains exist but are limited by the common factor.

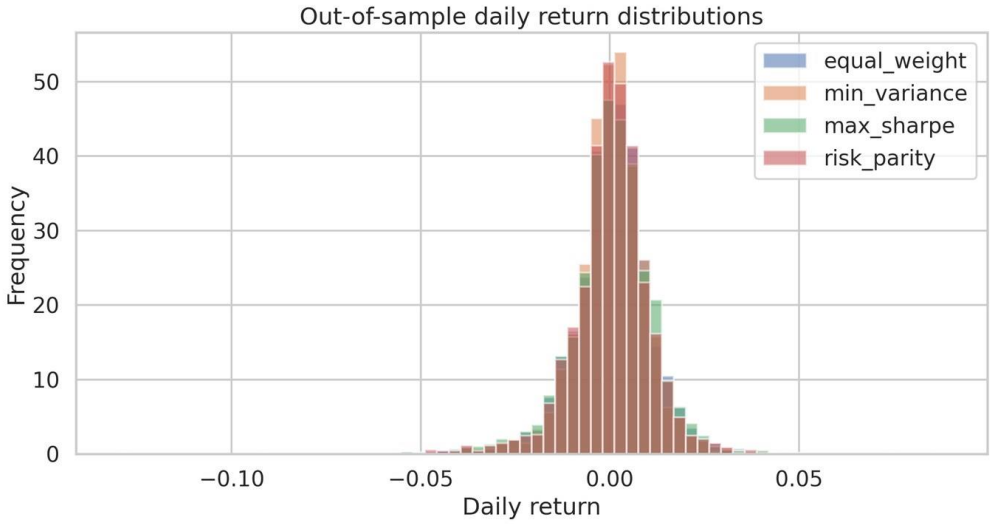


Figure 3. Out-of-sample daily return distributions of the four strategies.

Commentary. The four distributions are centred close to zero and overlap heavily, which is expected because every strategy is a long-only portfolio of the same 20 stocks. The Minimum Variance histogram is the most peaked, with more observations near zero and thinner shoulders, consistent with its lower volatility. Maximum Sharpe shows slightly more mass on the positive side. In all cases, a small number

of observations lie beyond $\pm 5\%$ (and, for the worst days, far beyond), so the tails are important and cannot be ignored by a variance-only analysis.

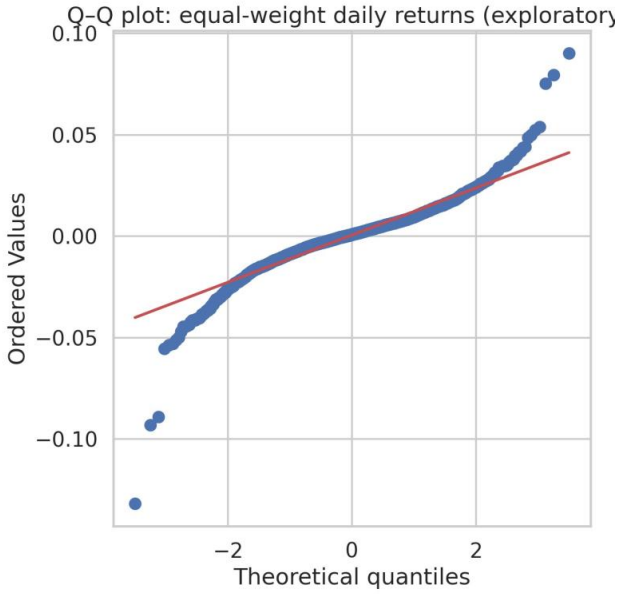


Figure 4. Normal Q–Q plot of equal-weight daily returns (exploratory).

Commentary. If returns were normally distributed, the points would lie along the red line. They follow it closely only in the centre; in both tails they depart from it, and the departure is stronger on the downside, where the worst daily observation is close to -13%. This is the classical signature of heavy tails (excess kurtosis) and negative skewness in the extreme tail. It justifies the use of tail-based risk measures (VaR, CVaR, maximum drawdown) and of HAC inference, rather than inference that assumes normality.

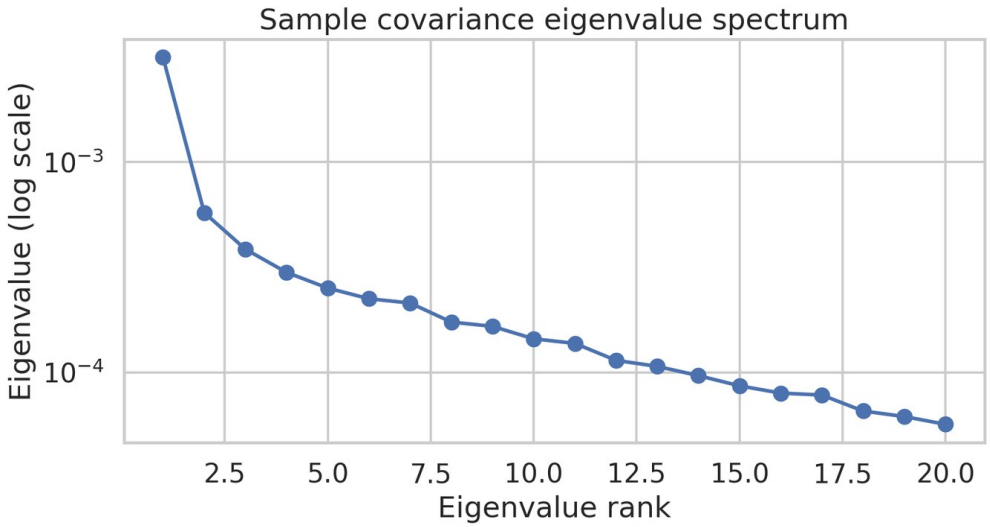


Figure 5. Eigenvalue spectrum of the sample covariance matrix (log scale).

Commentary. The largest eigenvalue, about 3×10^{-3}, is roughly five times larger than the second (about 6×10^{-4}), and the remaining eigenvalues decline gradually toward about 6×10^{-5}. The first eigenvalue corresponds to the common market factor, which drives most of the variance. The smallest eigenvalues describe low-variance directions that are estimated with the greatest relative error, and a

minimum-variance optimizer is attracted precisely to these directions. This is the main motivation for covariance shrinkage, tested in Section 13.

> **Takeaway.** The data warn of four problems: past returns are very unequal (Figure 1), one common factor limits diversification (Figures 2 and 5), tails are heavy (Figures 3 and 4), and the small eigenvalues are noisy (Figure 5). These become testable predictions: optimizers should look better in-sample, all strategies should share the same crashes, variance alone is not enough to describe risk, and strategies that depend most on estimation, especially Maximum Sharpe, are the most exposed to estimation error.

5. Portfolio Construction

Why this analysis. Section 4 described the inputs; this section shows what an optimizer promises when it uses them. Placing the four strategies on the in-sample efficient frontier gives the benchmark of what optimization is supposed to deliver. The gap between this in-sample promise and the out-of-sample delivery in Section 6 is the practical meaning of estimation risk.

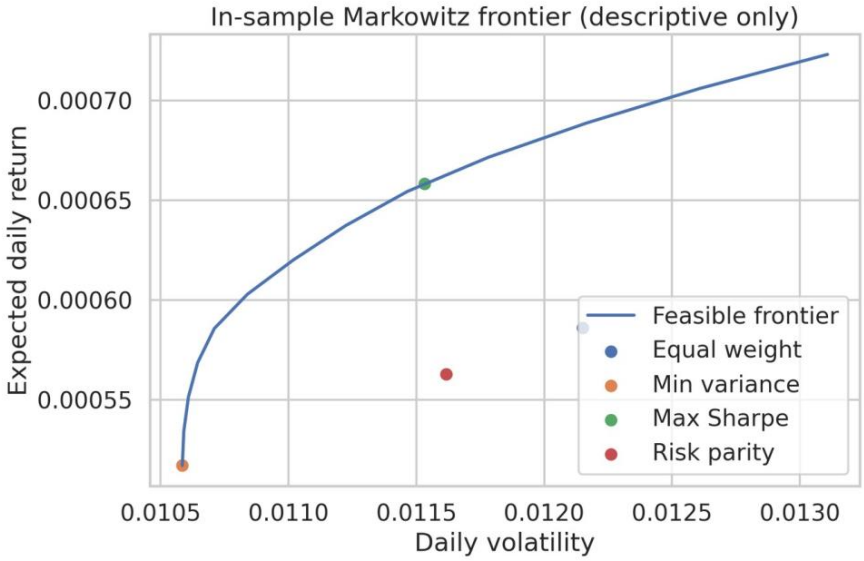


Figure 6. In-sample mean–variance efficient frontier with the four strategies (descriptive only).

Commentary. The curve is the in-sample feasible frontier under the long-only and 10% cap constraints. Minimum Variance sits at its leftmost point (daily volatility of about 0.0106). Maximum Sharpe lies on the frontier at a daily volatility of about 0.0115, which corresponds to roughly 18% annualized. Equal Weight (partly hidden behind the legend, around 0.0122 volatility and 0.00059 expected return) and Risk Parity (about 0.0116 and 0.00056) both lie clearly below the frontier. In-sample, therefore, the optimized portfolios dominate by construction. The figure is purely descriptive: the real test is whether this advantage survives when weights are chosen with past data and evaluated on future data, which is the subject of the next section.

> **Takeaway.** In-sample, optimization dominates Equal Weight by construction. This is the promise. Section 6 tests whether it is delivered when the weights are estimated with past data only.

6. Baseline Out-of-Sample Results

Why this analysis. This is the core test of the research question. Weights are estimated on past data only and evaluated on data the optimizer has never seen, so any advantage is genuinely out-of-sample and can be compared with the in-sample promise of Section 5.

Table 3 reports the walk-forward results over the 2,192 out-of-sample days, after the 10 basis point transaction cost.

Table 3. Out-of-sample performance and risk of the four strategies.

Strategy	Ann. return	Ann. vol.	Sharpe	Sortino	Max DD	Calmar	VaR 95%	CVaR 95%
Equal Weight	12.65%	18.82%	0.728	1.005	−38.10%	0.332	1.70%	2.88%
Minimum Variance	10.15%	16.37%	0.674	0.919	−34.93%	0.291	1.47%	2.49%
Maximum Sharpe	14.07%	18.14%	0.817	1.130	−35.11%	0.401	1.71%	2.76%
Risk Parity	11.97%	17.76%	0.726	0.997	−36.96%	0.324	1.61%	2.73%

Reading the table. Maximum Sharpe has the best return, Sharpe, Sortino and Calmar ratios, and a smaller maximum drawdown than Equal Weight (−35.11% against −38.10%). Minimum Variance achieves what it is designed to do: the lowest volatility, VaR and CVaR, but at the cost of the lowest return and the lowest Sharpe ratio. Risk Parity lies close to Equal Weight in Sharpe ratio (0.726 against 0.728) with slightly lower volatility. The differences between Equal Weight, Risk Parity and Maximum Sharpe are therefore modest in size, a point confirmed by the formal tests of Section 8.

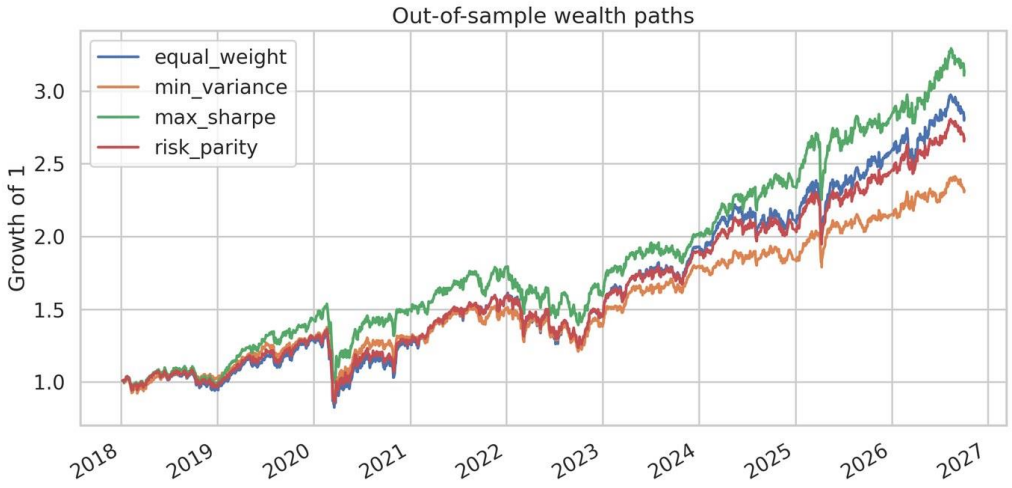


Figure 7. Out-of-sample cumulative wealth paths (growth of 1).

Commentary. All four strategies roughly double or triple the initial investment over the period. Maximum Sharpe (green) finishes highest, at a little above 3, followed by Equal Weight (about 2.8), Risk Parity (about 2.7) and Minimum Variance (about 2.3). The lead of Maximum Sharpe builds gradually from 2019–2020 and widens in 2025–2026, so it depends partly on the final part of the sample. The curves share the same dips, notably in early 2020 and during 2022, which confirms that the strategies are all dominated by the same market risk. Minimum Variance consistently ends below the others, which is the price paid for lower volatility.

Commentary. Drawdowns are strongly synchronized across strategies. The deepest episode is the early-2020 crash (−38.1% for Equal Weight and about −35% for the optimized portfolios), followed by a prolonged decline in 2022–2023 with drawdowns of roughly 20% to 25%, and a sharp but short fall in 2025. The optimizers reduce the depth of the 2020 drawdown only by about two to three percentage points. This shows that optimizing weights within a single long-only equity universe cannot protect against systematic shocks, because the common factor identified in Figure 5 dominates the risk.

Takeaway. The promise is only partly delivered. Maximum Sharpe leads on return, Sharpe, Sortino and Calmar ratios, but the differences are modest, and all strategies share the same crashes, as predicted in Section 4. Two questions follow: is this result stable over time (Section 7), and is it larger than chance (Section 8)?

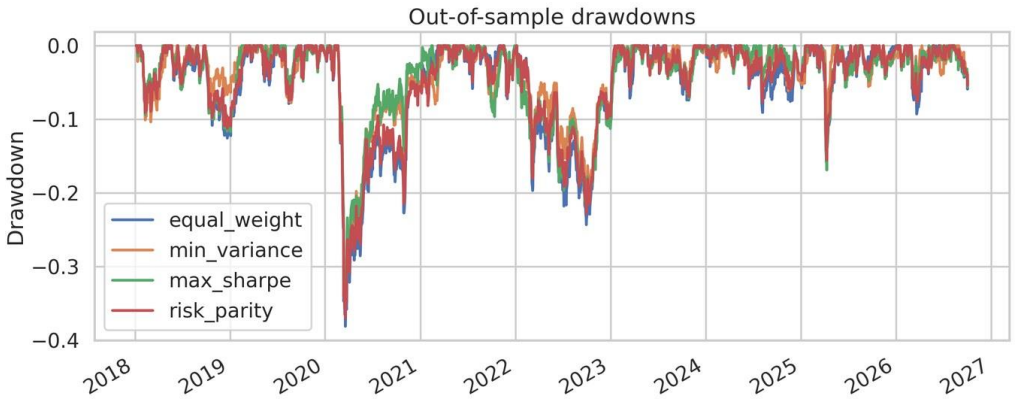


Figure 8. Out-of-sample drawdowns of the four strategies.

7. Dynamic Risk Analysis

Why this analysis. Section 6 gives one number per strategy for almost nine years. If risk and performance change a lot over time, a single number can be misleading, and a difference between two strategies may depend on the period. This section therefore looks at how risk and performance evolve, which also prepares the HAC inference (Section 8) and the regime analysis (Section 10).

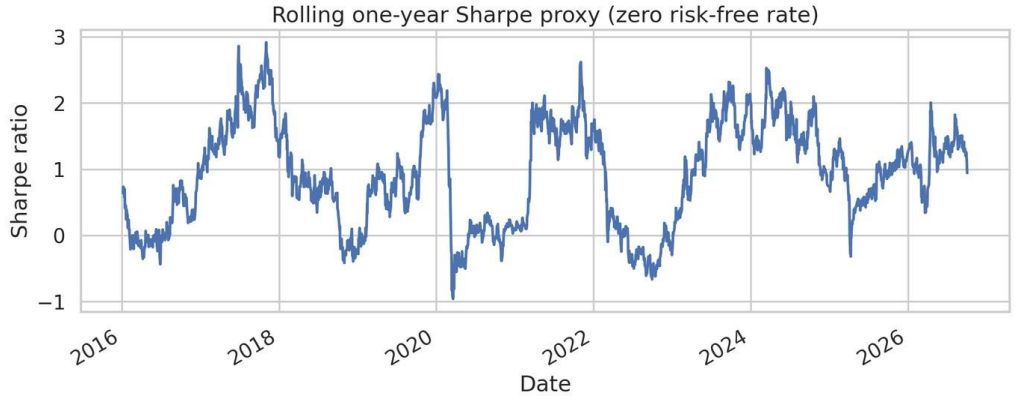


Figure 9. Rolling one-year Sharpe ratio proxy (zero risk-free rate).

Commentary. The rolling Sharpe ratio fluctuates between about -1 and 3. It is strongly positive in 2017, 2021 and 2023–2024, but turns negative in 2018, around the 2020 crash and again in 2022. Because the series begins in 2016, one year after the start of the data, it describes the whole sample and not only the out-of-sample period. Its main message is that any single full-sample Sharpe ratio summarizes a very unstable quantity: over one-year windows the ratio changes sign several times. A difference of 0.09 between two full-sample Sharpe ratios (0.817 against 0.728) is small relative to this variation.

Commentary. Volatility oscillates between calm periods of about 10% to 15% and sharp spikes: roughly 30% around 2015–2016, more than 50% at the peak of the 2020 crisis, about 33% in 2022 and about 25% in 2025. These spikes are short and clustered, a well-known feature of financial returns. Volatility clustering is the reason why constant-variance assumptions are inadequate and why the study uses HAC inference and a volatility-regime analysis.

Commentary. The ordering of the strategies is consistent with their design. Minimum Variance has the lowest volatility in most periods, and Equal Weight the highest during stress episodes, for instance at the peak of 2020, where it reaches about 54% against about 49% for Maximum Sharpe. The picture

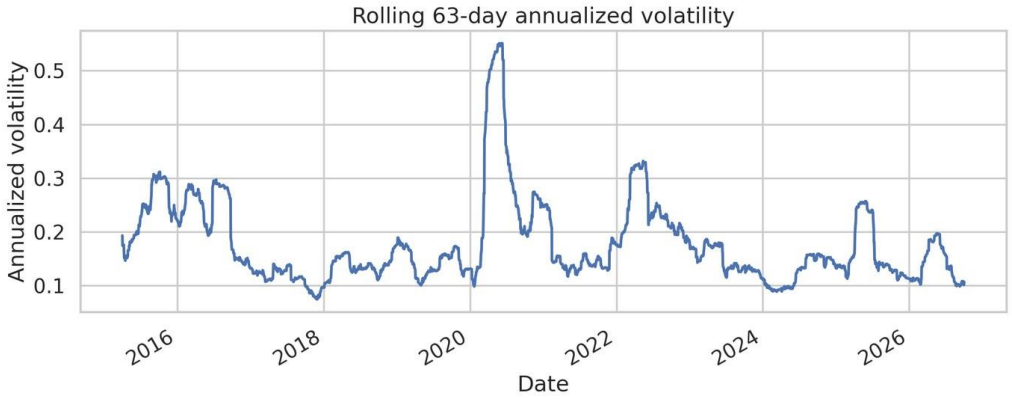


Figure 10. Rolling 63-day annualized volatility.



Figure 11. Rolling annualized volatility of the four strategies.

is not uniform, however: in the 2025 episode Maximum Sharpe is at least as volatile as Equal Weight. Therefore, the risk reduction offered by optimization is not guaranteed in every market environment. It depends on how well the covariance estimated in the past describes the future.

Takeaway. Performance and risk are highly unstable over time: rolling Sharpe ratios change sign repeatedly and volatility spikes in short bursts. A full-sample difference is therefore fragile, which calls for formal testing with HAC standard errors (Section 8) and for conditioning on market regimes (Section 10).

8. Statistical Inference

Why this analysis. Section 6 shows that Maximum Sharpe has a higher return and Sharpe ratio than Equal Weight, but a difference in averages is evidence only if it is large relative to the noise in the data. Section 7 showed that returns are volatile and clustered, so the noise is substantial. A formal test turns the descriptive comparison into inference, and this is the decisive step for the research question.

Table 4 presents the paired HAC test of the difference in mean daily return between each strategy and Equal Weight (Section 3.5).

Table 4. Paired HAC test of mean daily return differences relative to Equal Weight.

Comparison	Mean difference	HAC SE	t-statistic	p-value
Minimum Variance	-0.0001062	0.0000902	-1.178	0.239
Maximum Sharpe	$+0.0000445$	0.0000898	0.495	0.621
Risk Parity	-0.0000318	0.0000294	-1.082	0.279

Commentary. None of the three comparisons is statistically significant at conventional levels (all p-values are above 0.23). For Maximum Sharpe, the mean difference of $+0.0000445$ per day corresponds approximately to $+1.1$ percentage points per year (simple annualization by 252), but its standard error is about 0.0000898, or roughly 2.3 percentage points per year. The observed advantage is therefore about half of one standard error, well inside the range of sampling noise. Minimum Variance and Risk Parity have negative point estimates, but these are not significant either. Taken together, the data neither confirm that optimization outperforms Equal Weight nor show that it underperforms, and the test cannot reject the possibility that the observed differences are due to chance.

Takeaway. No strategy beats Equal Weight at a conventional significance level. For Maximum Sharpe, $p = 0.621$. The remaining question is whether its higher average return is simply compensation for more market risk (Section 9).

9. CAPM-Style Regression

Why this analysis. A strategy can earn more simply by taking more market risk. The regression separates the part of each strategy's return that comes from market exposure (beta) from the part that the market does not explain (alpha). It follows from Figure 5, where one common factor dominates the covariance structure, and it complements Section 8 by asking where the return difference comes from.

Table 5 reports the regression of each strategy's daily return on the European market proxy EXSA.DE, with a zero risk-free rate.

Commentary. The betas show the expected structure: Minimum Variance has the lowest market exposure (0.926), while Equal Weight has the highest (1.112), meaning that it is more sensitive than the market proxy. R^2 values between 0.82 and 0.90 indicate that most of each strategy's variance is explained by the market, which is consistent with the dominant first eigenvalue in Figure 5. Alphas are

Table 5. Market regression of the four strategies (daily alpha).

Strategy	Daily α	β	α t-stat	p-value	R^2
Equal Weight	0.0001729	1.112	2.182	0.0292	0.897
Minimum Variance	0.0001285	0.926	1.408	0.1591	0.821
Maximum Sharpe	0.0002430	1.038	2.771	0.0056	0.842
Risk Parity	0.0001615	1.050	2.199	0.0279	0.899

positive for all strategies. Maximum Sharpe has the largest and most significant alpha (about 6.1% per year, $p = 0.0056$), compared with about 4.4% for Equal Weight ($p = 0.029$), 4.1% for Risk Parity and 3.2% for Minimum Variance, which is not significant (annualized values are approximate, obtained by multiplying the daily alpha by 252).

These alphas should be interpreted with care. Equal Weight also has a significantly positive alpha, which suggests that much of it comes from the composition of the universe (stocks selected because they are large and successful today, see Section 17) and not from the optimization itself. The regression also uses a zero risk-free rate. The alpha difference between Maximum Sharpe and Equal Weight is of the order of 1.8 percentage points per year, which is consistent with the non-significant HAC result, although the difference in alphas is not tested directly.

Takeaway. Most of each strategy's variance is market-driven. Maximum Sharpe has the largest alpha, but Equal Weight also has a significant alpha, so alpha here mixes the effect of the stock universe with the effect of optimization. Section 10 asks when in time the advantage appears.

10. Volatility-Regime Analysis

Why this analysis. Section 7 showed that volatility comes in clusters. An average difference close to zero over the full sample could hide a pattern, for example optimization helping in calm markets and not in turbulent ones. The regime analysis checks whether the weak overall result conceals a systematic advantage under specific market conditions.

Days are classified into three regimes according to the rolling volatility of Equal Weight: the lowest 20% of volatility readings are labelled low-volatility, the highest 20% high-volatility, and the remaining 60% normal. Table 6 reports annualized performance conditional on each regime.

Commentary. The regimes behave very differently. In the high-volatility regime, annualized returns are very large (50% to 58%) with Sharpe ratios of about 1.6 to 1.7 for all strategies, with almost no difference between them. This should not be read as an ability to profit from crises: the regime is defined using volatility measured over the same window, and it includes the sharp rebounds that follow selloffs, so the label is descriptive and not a tradable signal. In the low-volatility regime, Maximum Sharpe has the best Sharpe ratio (1.892 against 1.394 for Equal Weight), and Minimum Variance also improves on Equal Weight (1.672) with much lower volatility. In the normal regime, which covers most days, Sharpe ratios are low for everyone (0.23 to 0.41), and Maximum Sharpe is only slightly better than Equal Weight (0.411 against 0.360). Optimization therefore seems to help most in calm markets, where covariance estimates are most reliable, and it adds little elsewhere.

Takeaway. The advantage of Maximum Sharpe is concentrated in calm markets (Sharpe 1.892 against 1.394) and small in normal and turbulent ones. The weak average result of Section 8 therefore hides an uneven pattern, but not a robust, across-the-board advantage. The next step is to examine what it costs to obtain this performance.

Table 6. Strategy performance by volatility regime.

Regime	Strategy	Return	Volatility	Sharpe
High	Equal Weight	58.36%	35.13%	1.661
High	Minimum Variance	50.43%	30.35%	1.662
High	Maximum Sharpe	52.14%	31.01%	1.681
High	Risk Parity	54.31%	33.26%	1.633
Low	Equal Weight	15.91%	11.41%	1.394
Low	Minimum Variance	16.64%	9.95%	1.672
Low	Maximum Sharpe	22.30%	11.79%	1.892
Low	Risk Parity	17.05%	10.63%	1.605
Normal	Equal Weight	6.08%	16.90%	0.360
Normal	Minimum Variance	3.31%	14.75%	0.225
Normal	Maximum Sharpe	6.98%	16.97%	0.411
Normal	Risk Parity	5.27%	15.93%	0.331

11. Concentration and Implementation

Why this analysis. Sections 6 to 10 judged the strategies by their returns. But a strategy is only useful if it can be implemented, and a high return achieved with a concentrated portfolio is a different thing from the same return achieved with a diversified one. This section measures concentration, and Section 12 measures trading, so that the cost of obtaining the performance can be weighed against the performance itself.

Table 7. Concentration statistics of the four strategies.

Strategy	Avg. max weight	Avg. active positions	Min. active	Effective assets	Cap rate
Equal Weight	5.00%	20.0	20	20.00	0%
Maximum Sharpe	10.00%	11.0	10	10.38	100%
Minimum Variance	10.00%	12.9	10	10.97	100%
Risk Parity	8.84%	20.0	20	18.55	8.57%

Commentary. The two strategies that rely most on estimation are also the most concentrated. Maximum Sharpe holds only about 11 of the 20 stocks on average and has about 10.4 effective assets, which is roughly half of Equal Weight's 20. Its largest position equals the 10% cap at every rebalance (cap rate of 100%), and the same is true for Minimum Variance. In other words, the cap is binding throughout and acts as the main safeguard against extreme weights: without it, the optimizers would probably concentrate even more. Risk Parity remains broadly diversified (about 18.6 effective assets) and reaches the cap only occasionally. This analysis shows that the apparent performance of Maximum Sharpe comes from a considerably more concentrated bet.

Takeaway. Maximum Sharpe obtains its results with a portfolio that is about half as diversified as Equal Weight and that hits the 10% cap at every rebalance. Section 12 shows how much it also trades.

12. Turnover and Transaction Costs

Why this analysis. Concentration and turnover are linked: an optimizer whose weights react to noisy estimates changes its positions frequently, and every change has a cost. This section quantifies how much each strategy trades and what this costs under the assumed 10 basis points.

Table 8. Turnover and ongoing transaction costs (10 bps per unit of turnover).

Strategy	Mean ongoing turnover	Total ongoing turnover	Ongoing TC
Equal Weight	7.90%	2.687	0.269%
Maximum Sharpe	26.69%	9.074	0.907%
Minimum Variance	14.34%	4.876	0.488%
Risk Parity	7.91%	2.688	0.269%

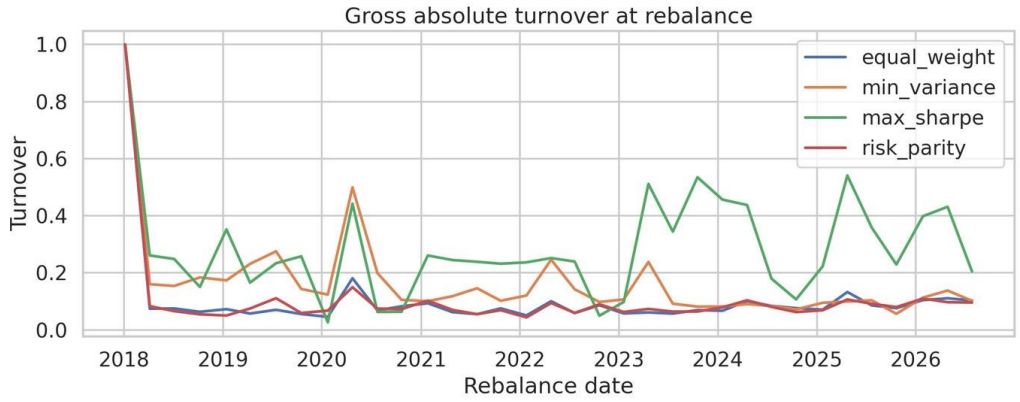


Figure 12. Gross absolute turnover at each rebalancing date.

Commentary. The first rebalance in January 2018 has a turnover of 100% for every strategy, since the portfolio is built from cash. Afterwards, Equal Weight and Risk Parity keep turnover at about 5% to 13% per rebalance, which comes only from price drift. Maximum Sharpe is much more active: its mean turnover of 26.69% is about 3.4 times that of Equal Weight, with peaks of around 50% in 2023 and 2025. Minimum Variance lies in between, with notable peaks around 2020 and 2023. Cumulative ongoing costs are 0.907% for Maximum Sharpe against 0.269% for Equal Weight (the sum of 0.1% of turnover over the whole sample, equivalent to roughly 0.1 percentage points of annual return for Maximum Sharpe). At 10 basis points and quarterly rebalancing, costs are small compared with returns, but they are more than three times higher for the strategy that is supposed to be superior.

Takeaway. Costs are small in absolute terms but more than three times those of Equal Weight. Section 14 checks whether the advantage survives higher costs.

13. Estimation Robustness

Why this analysis. The research question explicitly asks about estimation uncertainty, and this section tests it directly. If the results change when the inputs are made more robust (by shrinking the covariance matrix, as motivated by Figure 5, and by shrinking expected returns), the original performance partly reflects estimation noise. If they do not change, the ranking is not an artifact of one estimator.

Three specifications are compared for Maximum Sharpe: (i) the sample covariance with historical mean returns; (ii) the Ledoit–Wolf shrinkage covariance with historical mean returns; and (iii) the

Ledoit–Wolf covariance with expected returns shrunk by 50% toward their cross-sectional average:

$$\tilde{\mu} = 0.5\,\hat{\mu} + 0.5\,\bar{\mu}\mathbf{1} \tag{31}$$

Table 9. Robustness of Maximum Sharpe to the estimation method.

Specification	MS return	MS volatility	MS Sharpe
Sample covariance + historical mean	14.07%	18.14%	0.817
Ledoit–Wolf + historical mean	14.08%	18.14%	0.817
Ledoit–Wolf + mean shrinkage	13.58%	17.66%	0.810

Commentary. Results are almost unchanged. Replacing the sample covariance by the Ledoit–Wolf estimator leaves the return, volatility and Sharpe ratio virtually identical, and shrinking the expected returns reduces the return by about half a percentage point and the volatility by a similar amount, leaving the Sharpe ratio at 0.810. A plausible explanation is that the constraints (long-only and 10% cap) already regularize the problem, and that with 20 assets and 756 observations, the sample covariance is reasonably well conditioned. The qualitative ranking is therefore not caused by one particular estimator. The conclusion applies only to the specifications tested.

Takeaway. The ranking is not an artifact of the covariance estimator or of the raw mean estimates: results barely move under Ledoit–Wolf shrinkage or mean shrinkage, probably because the long-only and cap constraints already regularize the problem. Robust, however, does not mean significant: Section 8 still applies.

14. Transaction-Cost Sensitivity

Why this analysis. The last component of the research question is transaction costs. Section 12 computed costs at one level (10 basis points); here the assumption is varied from zero to 50 basis points to check whether the advantage of Maximum Sharpe depends on it.

Table 10. Sensitivity of the Maximum Sharpe (MS) versus Equal Weight (EW) comparison to transaction costs.

Cost	EW return	MS return	EW Sharpe	MS Sharpe	MS − EW return
0 bps	12.70%	14.20%	0.730	0.823	+1.50 pp
10 bps	12.65%	14.07%	0.728	0.817	+1.41 pp
25 bps	12.58%	13.87%	0.724	0.808	+1.29 pp
50 bps	12.46%	13.54%	0.719	0.792	+1.08 pp

Commentary. As costs rise from 0 to 50 basis points, the return advantage of Maximum Sharpe over Equal Weight falls from 1.50 to 1.08 percentage points, and its Sharpe ratio remains above Equal Weight's at every cost level (0.792 against 0.719 at 50 bps). The cost assumption alone, therefore, does not eliminate the point-estimate advantage, mainly because rebalancing is only quarterly. However, since the underlying difference is not statistically significant (Section 8), the cost analysis changes the size of a statistically uncertain advantage but cannot make it significant.

Takeaway. The point-estimate advantage persists even at 50 basis points, but remains statistically unproven. The evidence from all analyses is now combined in Sections 15 and 16 to answer the research question.

15. Discussion

The results can be read at three levels. Descriptively, optimization helps: Maximum Sharpe has the best return, Sharpe, Sortino and Calmar ratios, a lower maximum drawdown than Equal Weight, and the largest market-regression alpha, and this survives reasonable increases in transaction costs and changes in the covariance estimator. Statistically, the evidence is weak: the paired HAC test gives $p = 0.621$, and the annualized advantage of about one percentage point is only about half of its standard error. Practically, the advantage comes at a price. Maximum Sharpe hits the 10% cap at every rebalance, holds about half as many effective assets as Equal Weight and trades roughly 3.4 times more.

The other strategies illustrate the trade-offs. Minimum Variance gives the clearest risk-control profile (lowest volatility, VaR and CVaR) at the cost of the lowest return. Risk Parity offers broad diversification and low turnover, with performance almost identical to Equal Weight. Equal Weight, with no estimation at all, proves difficult to dominate statistically. This is in the spirit of DeMiguel, Garlappi and Uppal (2009): sophistication in the objective function does not automatically produce better out-of-sample results when the inputs are noisy.

The regime analysis adds a nuance: the benefits of optimization seem concentrated in calm markets, where past covariances are most informative, while in normal and turbulent regimes the differences between strategies are small. Finally, the dynamic analysis shows how unstable the underlying performance is. Rolling Sharpe ratios change sign several times, so a single full-sample statistic should not be over-interpreted.

16. Answer to the Research Question

The research question asked whether sophisticated portfolio optimization provides superior risk-adjusted performance compared with a simple equally weighted portfolio once estimation uncertainty, downside risk, transaction costs and changing market conditions are taken into account. Table 11 breaks the question into its components and compares Maximum Sharpe, the most sophisticated strategy and the descriptive leader, with Equal Weight (EW).

Table 11. Evidence for each component of the research question (Maximum Sharpe, MS, versus Equal Weight, EW).

Component	Evidence	Verdict
Risk-adjusted performance	Sharpe 0.817 vs 0.728; Sortino 1.130 vs 1.005; Calmar 0.401 vs 0.332	Better in the sample
Statistical reliability	Mean daily difference $+0.0000445$; HAC $t = 0.495$, $p = 0.621$	Not significant
Estimation uncertainty	Sharpe 0.817 / 0.817 / 0.810 under sample, Ledoit–Wolf and mean shrinkage; but the cap binds at every rebalance and MS holds 10.4 effective assets against 20	Robust to the estimator, but exposed
Downside risk	Max drawdown -35.11% vs -38.10%; CVaR 2.76% vs 2.88%; VaR 1.71% vs 1.70%	Slightly better in the tails, not decisive
Transaction costs	Mean turnover 26.69% vs 7.90% per rebalance; return advantage still $+1.08$ pp at 50 bps	Survives costs as a point estimate
Changing market conditions	Sharpe 1.892 vs 1.394 (low volatility), 0.411 vs 0.360 (normal), 1.681 vs 1.661 (high)	Concentrated in calm markets

Final answer. Not demonstrably. In this experiment, sophisticated optimization, and Maximum Sharpe in particular, produced better risk-adjusted performance than Equal Weight on paper, and this

advantage survives reasonable transaction costs and covariance shrinkage. However, the advantage cannot be distinguished from zero statistically ($p = 0.621$), it is obtained with a portfolio that is about half as diversified and trades about 3.4 times more, it appears mainly in calm markets, and it rests on a single sample of 20 large-cap stocks that are still listed today. The honest answer is therefore conditional: optimization may improve risk-adjusted performance, but this study provides no reliable evidence that it does so systematically, and the equally weighted portfolio remains a strong benchmark that is hard to beat once estimation uncertainty and implementation realities are respected.

What would change this answer. A longer or broader sample, a survivorship-free universe, or a direct test of Sharpe-ratio differences that found a significant advantage would support the stronger claim. A significantly negative result would support the opposite. Until then, the practical lesson is that the value of optimization depends less on the sophistication of the objective function than on the quality of the inputs, the constraints and the costs of trading.

17. Limitations

- **Survivorship and selection bias.** The universe consists of stocks that are large and listed today, which favours all strategies, including Equal Weight, and can inflate alphas.
- **Data quality.** Yahoo Finance data are not equivalent to institutional market-data infrastructure and may contain adjustments or errors.
- **Modelled costs.** Transaction costs are assumed at a fixed rate per unit of turnover; actual execution costs, market impact and taxes are not observed.
- **Regression design.** The CAPM-style regression uses a zero risk-free rate and a single European index proxy.
- **Limited robustness.** Robustness is restricted to the specifications explicitly tested; other training windows, rebalancing frequencies, caps or universes are not examined.
- **Scope of inference.** The HAC test concerns the mean daily return difference. It does not directly test differences in Sharpe ratios, and no correction is applied for comparing several strategies at once.
- **Single universe and period.** Results depend on 20 large-cap European stocks and on a period that includes unusual events (the 2020 crash, the 2022 shock); they may not generalize.
- **Nature of the evidence.** The study is descriptive and predictive rather than causal, and historical performance does not guarantee future performance.

18. Directions for Future Research

Several extensions would strengthen the analysis: formal tests and bootstrap confidence intervals for differences in Sharpe, Sortino and drawdown measures; a survivorship-free universe built from point-in-time index constituents; alternative training windows, rebalancing frequencies and weight caps; additional estimators such as factor-model covariances, Black–Litterman views or CVaR-based optimization; and a wider asset universe including other regions or asset classes.

19. Conclusion

In this historical walk-forward experiment on 20 European equities, portfolio optimization produces stronger descriptive risk-adjusted performance than Equal Weight, and Maximum Sharpe is the descrip-

tive leader. Its advantage is not statistically significant in the paired HAC comparison ($p = 0.621$), it is accompanied by high concentration and turnover, and it depends on the estimation inputs and constraints.

The evidence therefore does not support the stronger claim that sophisticated optimization systematically outperforms a simple benchmark. Instead, it shows that optimization can improve historical outcomes while introducing concentration, estimation sensitivity and implementation costs. The value of portfolio optimization depends not only on the theoretical objective function but on the quality of the estimated inputs, the constraints, the turnover, and the statistical uncertainty surrounding the observed performance difference.

20. Code Availability and Reproducibility

The entire analysis is implemented in Python and is available in a public GitHub repository named *portfolio-optimization-under-uncertainty*:
https://github.com/PinkMathJakim/portfolio-optimization-under-uncertainty. The code is designed to run in Google Colab, so that results can be reproduced without any local installation: the dependencies are listed in the repository's requirements file, and market data are downloaded automatically through yfinance.

The project is organized into ten sequential notebooks covering data collection and cleaning, exploratory analysis, return and covariance analysis, Markowitz optimization, risk parity, covariance shrinkage, Monte Carlo simulation, rolling backtest, econometric analysis and robustness analysis. Reusable logic is placed in a source package (data handling, returns, covariance estimation, optimization, risk measures, simulation, backtesting, econometrics and visualization), accompanied by unit tests for the return, risk, optimization and backtest modules, and an interactive Streamlit application. The final PDF report is stored in the repository.

References

DeMiguel, V., Garlappi, L., & Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies*, 22(5), 1915–1953. https://doi.org/10.1093/rfs/hhm075

Ledoit, O., & Wolf, M. (2004). Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management*, 30(4), 110–119. https://doi.org/10.3905/jpm.2004.110

Maillard, S., Roncalli, T., & Teïletche, J. (2010). The properties of equally weighted risk contribution portfolios. *Journal of Portfolio Management*, 36(4), 60–70. https://doi.org/10.3905/jpm.2010.36.4.060

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91. https://doi.org/10.2307/2975974

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. https://doi.org/10.2307/1913610

Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2, 21–42. https://doi.org/10.21314/JOR.2000.038

Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3), 425–442. https://doi.org/10.1111/j.1540-6261.1964.tb02865.x